RUIYANG WANG

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Robotics & LLM Research | Embodied AI | Multi-Robot Autonomy

Education

- **Duke University** — Durham, NC
  Ph.D. in Electrical and Computer Engineering — *Aug 2024 – Apr 2028 (Expected)*
- **University of Michigan, Ann Arbor** — Ann Arbor, MI
  M.S. in Robotics; B.S. in Mechanical Engineering — *Aug 2018 – Dec 2023*

Research Experience

- **Duke University — Cyber-Physical Systems Lab** — Durham, NC
  Ph.D. Researcher | Advisor: Prof. Miroslav Pajic — *Aug 2024 – Present*
  - Developed **multi-robot autonomy** methods using LLM/VLMs for exploration and search coordination, **graph attention networks** for trust inference, and **neural-symbolic reasoning** for deadlock resolution.
  - Building **graph-based planning harnesses** to verify and repair LLM/VLM-generated robot action plans, improve **long-horizon task completion**, and check task and safety constraints before execution.

Selected Publications & Manuscripts

[1] **GAVEL: Graph World Models for Verified and Efficient Long-Horizon LLM Task Planning**
Ruiyang Wang, Hao-Lun Hsu, Swarajh Mehta, Jiwoo Kim, Zhihao Dou, Miroslav Pajic
Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027

- Designed graph-based plan verification, repair, and belief-aware task ordering; raised Qwen3-8B success from **41.2% to 91.8%** on single tasks and **19.9% to 92.6%** on multi-task instructions in OmniGibson with BEHAVIOR-1K scenes.

[2] **COMRES-VLM: Coordinated Multi-Robot Exploration and Search using Vision Language Models**
Ruiyang Wang, Hao-Lun Hsu, David Hunt, Jiwoo Kim, Shaocheng Luo, Miroslav Pajic
Submitted to IEEE Robotics and Automation Letters (RA-L)

- Developed **regional VLM planning** with **natural-language hints**; cut search time by **25%** versus Hungarian with priors and **33%** versus Co-NavGPT across **40 maps**, including planning delays; evaluated on two **Unitree Go2** robots.

[3] **NeST-Bayes — Who to Trust in Multi-Robot Systems: Joint Robot and Object Hypothesis Trust Inference via Neural-Symbolic Cross-Validation**
Ruiyang Wang, Hao-Lun Hsu, Haocheng Meng, Songqiao Xie, Miroslav Pajic
Submitted to IEEE Robotics and Automation Letters (RA-L)

- Developed **joint robot and object trust inference** from cross-robot report consistency using **heterogeneous graph attention** and temporal Beta updates; achieved **98.8%/98.3% robot/object precision** against aggressive optimized attacks and **92.5%/97.3%** on simulated Spot robots in Webots without retraining.

[4] **SAGR — Semantic Area Graph Reasoning for Multi-Robot Language-Guided Search**
Ruiyang Wang, Hao-Lun Hsu, Jiwoo Kim, Miroslav Pajic
arXiv preprint

- Designed **semantic area graphs** for LLM coordination with prompts of **fewer than 500 tokens**; reduced large-environment search time by **18.8%** versus AEP+DVC using GPT-4o, evaluated on 100 HM3D scenarios.

[5] **NSDR — Neuro-Symbolic Deadlock Resolution in Multi-Robot Systems**
Ruiyang Wang, Bowen He, Miroslav Pajic
Accepted at Learning for Dynamics and Control (L4DC), 2025

- Learned which robots should yield using **Neural Logic Machines** and MPC feasibility checks; generalized from **2–5 to 8 and 10 robots**, reducing average arrival time by **23% versus greedy** in the 10-robot test.

Technical Skills

Programming Languages: Python, C++, MATLAB, R, Julia
ML & LLMs: PyTorch, Hugging Face Transformers, scikit-learn, SFT, LoRA, RAG, vLLM, Weights & Biases
LLM Agents: Planning harnesses, structured action generation, plan verification and repair
Systems & Cloud: Linux (Ubuntu), Git, Docker, ROS2, AWS, Google Cloud
Simulation Tools: OmniGibson, Webots, PyBullet, HM3D, OpenAI Gym, BEHAVIOR-1K